

Gaze-Contingent Guidance in Mixed-Reality Visual Search

Draft content organized for a UCF HRP-503 protocol application

FIRST-PASS DRAFT

Not IRB-approved. Replace all bracketed open decisions and transfer final content into the current official UCF forms before submission.

1. Administrative Summary

Field	Draft entry
Principal investigator	[OPEN DECISION: name, department, email, phone]
Faculty advisor	[OPEN DECISION: name, department, email]
Study site	[OPEN DECISION: approved UCF laboratory room]
Population	Adults 18 or older who can safely use an MR headset and complete an English-language visual-search task.
Local enrollment	[OPEN DECISION: power/precision analysis and pilot allowance]
Session length	Approximately 45-60 minutes; replace with pilot timing.
Compensation	[OPEN DECISION: amount, form, and partial-payment policy]
Current executable design	One within-participant run of 14 rounds alternating gaze-unaware and gaze-aware spoken guidance.
Future design	A 2 x 2 gaze-awareness by voice-similarity design is proposed but is not implemented.

2. Background and Purpose

Mixed-reality visual search requires people to distribute attention across spatially arranged objects while coordinating eye gaze, head pose, target memory, and selection. The study evaluates whether spoken guidance that can use gaze-derived interaction state improves search performance or experience relative to guidance that does not use gaze information.

Prior visual-search and gaze research motivates eye calibration, controlled stimuli, explicit timing boundaries, workload measurement, and careful preprocessing. The study adapts those practices but does not treat thresholds from other headsets as validated for the HTC Vive Focus Vision.

Primary research question: Does gaze-aware spoken guidance change search completion time relative to gaze-unaware spoken guidance?

Secondary questions: Do conditions differ in wrong captures, first-try success, subjective workload, perceived helpfulness, trust, reliance, distraction, intrusiveness, or gaze-derived exploratory behavior?

3. Design and Conditions

CURRENT DESIGN BOUNDARY

The IRB submission must select one executable design. The present build is the two-condition 14-round study. The voice-similarity factor should remain out of scope unless it is implemented, separately consented, and fully specified before submission.

Condition	Information available	Participant experience
Gaze-unaware	Task state and elapsed time; no current gaze/hover evidence.	Generic encouragement delivered by the same audio path.
Gaze-aware	Task state plus current/recent gaze-hover, shelf, scanning, and target-proximity evidence.	Spoken feedback may reflect whether search appears off-target, on-track, or near the target.

The causal comparison requires matched hint opportunities, timing, audio quality, voice identity, and semantic usefulness. The current implementation uses different default delays/intervals and therefore requires correction or a narrower pilot interpretation.

4. Participants

Inclusion criteria

- Age 18 or older.
- Able to understand the approved English consent and instructions.
- Able to wear the HTC Vive Focus Vision and view the search display with normal or corrected-to-normal vision.
- Able to distinguish the task colors or pass the selected color-vision screen.
- Able to hear spoken guidance at a comfortable level.
- Willing to permit coded gaze, interaction, task-performance, and questionnaire data collection.

Exclusion or stopping criteria

- Current dizziness, nausea, severe headache, unusual fatigue, or condition making headset use unsafe.
- History of severe simulator sickness, relevant vestibular disorder, or visually triggered seizure risk.
- Vision, color discrimination, or hearing limitation that prevents task completion after reasonable correction.
- Eye-tracking calibration that remains unusable after refitting and the approved retry procedure.
- Participant request to stop, significant discomfort, unsafe tracking, or unrecoverable hardware/software failure.

Glasses, contact lenses, eye makeup, prior XR experience, and prior eye-tracking experience should be recorded as descriptors unless piloting demonstrates a specific safety or data-quality reason for exclusion.

5. Recruitment and Consent

Recruitment may use approved email, flyer, participant-pool, or word-of-mouth methods. Materials will describe mixed-reality headset use, eye tracking, spoken guidance, questionnaires, session duration, eligibility, compensation, and voluntary participation without describing one condition as expected to be superior.

A trained researcher will conduct consent before research data collection, review procedures and risks, explain gaze-data collection and confidentiality, answer questions, and confirm that participation may be paused or

stopped without penalty. Written versus verbal consent and any waiver of documentation remain open decisions for the PI and IRB.

6. Study Procedure

1. Confirm minimal eligibility and review consent.
2. Assign a coded participant ID; keep scheduling and compensation records separate.
3. Collect approved baseline demographics, XR experience, and baseline simulator-sickness ratings.
4. Fit the Vive Focus Vision and complete vendor eye-tracking calibration.
5. Validate gaze across the intended search area; refit/recalibrate according to the approved rule.
6. Complete practice using targets and layouts excluded from analyzed trials.
7. Start the MR task and present the target during a fixation-cross transition.
8. For each round, display 56 objects, allow gaze-dwell selection, provide condition-dependent spoken guidance, and continue after wrong captures until the correct target is selected.
9. Provide breaks at the approved intervals and whenever requested.
10. Collect raw NASA-TLX and the guidance manipulation check after each condition block if a block design is implemented.
11. After the final task, collect post-SSQ, overall SUS, comparative guidance/privacy items, and a short interview.
12. Remove the headset, check participant comfort, debrief, confirm compensation, and document deviations.

7. Apparatus and Data

Source	Recorded data	Current status
gaze_log.csv	Rendered-frame timestamp, gaze origin/direction/rotation, available per-eye position/openness/fixation point, hovered object, target match, dwell progress, task state, blink estimate.	Implemented; not a confirmed native-rate stream and lacks complete validity/calibration metadata.
trial_events.csv	Game/objective events, correct/wrong captures, hover-derived episode boundaries, object metadata, dwell duration.	Implemented; existing fixation/saccade names are interaction proxies.
trial_summary.json	Per-round timing, wrong captures, first-try success, target/distractor hover summaries, blink summaries.	Implemented; some timing boundaries require correction.
nasa_tlx.csv	Six 0-100 raw NASA-TLX subscales and aggregate raw score.	Implemented once after the run; condition-specific collection requires runtime changes.
Study questionnaires	Demographics, XR experience, SSQ, SUS, guidance manipulation checks, privacy/trust, interview.	Protocol proposal; instrument set requires PI/IRB approval.

8. Measures and Analysis

Construct	Operational definition	Role and limitation
Search time	Seconds from dedicated search_onset/display-ready event to correct dwell capture.	Proposed primary outcome; current start boundary must be fixed.

Construct	Operational definition	Role and limitation
Selection accuracy	First-attempt correctness and number of wrong captures per round.	Confirmatory or secondary performance outcome.
Subjective workload	Six raw NASA-TLX subscales, each 0-100, after each condition block.	Primary workload measure; cannot compare conditions from one end-of-run rating.
Guidance manipulation	Post-block ratings of responsiveness to looking/search behavior and perceived timing.	Manipulation check.
Hover allocation	Target/distractor hover duration, hover episodes, revisits, and target-entry latency.	Exploratory interaction proxy; not a validated fixation measure.
Eye behavior	Offline gaze angular velocity, dispersion, and pilot-validated fixation/saccade features.	Exploratory until sampling, quality, and classifier validation are complete.
Usability and safety	SUS total score and pre/post SSQ symptom scores.	Secondary system and safety outcomes.

Repeated observations should be analyzed with models that account for participant, round/order, target identity, and condition. The final plan must predeclare the primary outcome, multiplicity strategy, missing-data rules, gaze-quality exclusions, and any participant/round exclusions.

9. Risks and Mitigation

Risk	Mitigation
Eye strain, headache, fatigue, headset discomfort	Fit headset carefully; provide breaks; stop on request or significant symptoms.
Simulator/MR sickness or disorientation	Stable passthrough scene, no artificial locomotion, pre/post SSQ, clear physical area, seated option if approved.
Frustration or embarrassment about performance	Neutral instructions and feedback; explain that wrong selections are expected; permit skipping optional questions.
Gaze-data privacy	Coded IDs, data minimization, restricted approved storage, separate linkage data, aggregate reporting.
Audio/network interruption	Preflight test, matched fallback behavior, deviation logging, no pressure to repeat after discomfort.
Future voice cloning	Out of current scope unless separately consented with provider, retention, deletion, and reuse limits.

10. Data Management

Research files will use coded participant IDs. Names, contact details, consent records, scheduling information, and compensation records will be stored separately from task data. Files will be transferred from the headset/study computer to [OPEN DECISION: approved UCF-managed storage], verified, and local working copies removed according to the approved procedure. Access, encryption, retention duration, deletion, withdrawal cutoff, and de-identified sharing language remain open decisions.

Gaze telemetry is sensitive behavioral data. It will not be described as diagnostic, and reports will use aggregate statistics or de-identified examples.

11. Required Pre-Submission Resolutions

- Select the executable study design and remove unsupported voice-factor language if out of scope.
- Match hint opportunities/timing/content and add a complete hint audit trail.
- Add a valid search_onset event.
- Counterbalance condition and target/layout order.
- Select sample size using an a priori power/precision analysis.
- Finalize compensation, site, study duration, and stopping rules.
- Finalize calibration validation, achieved-sampling, validity, missingness, and exclusion rules.
- Implement condition-specific NASA-TLX or limit workload claims.
- Finalize data storage, retention, deletion, access, sharing, and withdrawal language.
- Have the PI/faculty advisor verify instrument permissions, scoring, and current UCF template requirements.